Advanced Programming Practice · 21CSC203P

Unit 1 · Session 5 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 5 : SLO – 2 | Compare Paradigms

Activity (Solutions)

• 1) OOP vs functional in Java — OOP: objects carry mutable state and behavior; class Counter { int n=0; void inc(){ n++; } }. Functional: stateless pure transformations; IntUnaryOperator inc = n -> n + 1; list.stream().map(inc).forEach(...). OOP emphasizes encapsulation and identity; functional emphasizes immutability and composition.

• 2) Same problem (sum of even numbers) — Procedural: int s=0; for(int x: arr) if(x%2==0) s+=x; OO/functional style: int s = Arrays.stream(arr).filter(x -> x%2==0).sum(); — the first states how step-by-step; the second declares what to compute.

• 3) Declarative strengths: concise, readable, closer to problem domain, easier parallelization. Weaknesses: less control over performance, harder debugging of hidden control flow. Imperative strengths: full control, predictable performance, maps to hardware. Weaknesses: verbose, error-prone state management.